THE AFFINE ETALE SITE OF AN AFFINE FORMAL ALGEBRAIC SPACE IS THE CATEGORY OF REPRESENTABLE ETALE AFFINE FORMAL ALGEBRAIC SPACES OVER IT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The last sentence of the proof is the author's 'We omit the verification that the constructions are mutually inverse to each other.' It is reproduced verbatim and recorded as an obligation of the card. Both functors and their required properties are established in the span, so the omission concerns only the round-trip verification.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A representable etale morphism from an affine formal algebraic space to an affine formal algebraic space is representable by schemes and affine. Same chapter, Lemma lemma-affine-representable-by-algebraic-spaces, invoked at proof lines 7493-7496. Makes the fibre product with the reduction an affine scheme.
- Affine formal algebraic spaces as filtered colimits of affine schemes, with the reduction as a thickening of each stage. Same chapter, Definition definition-affine-formal-algebraic-space, invoked at proof lines 7504-7505. Supplies the system along which the quasi-inverse is built.
- Etale affine schemes over a thickening agree with those over the reduced scheme. More on Algebra, Lemma more-algebra-lemma-locally-nilpotent-henselian, invoked at proof lines 7508-7512. Extends an etale morphism over the reduction to a compatible system over all stages.

Source reference. Original lemma 0AKN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4209–4209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0ALI: sources/stacks-chow/tags.txt, source lines 13371–13371. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let X be an affine formal algebraic space over S. Then $X_{affine,étale}$ is equivalent to the category whose objects are morphisms $\varphi : U \to X$ of formal algebraic spaces such that*

(1) *U is an affine formal algebraic space,*
(2) *φ is representable by algebraic spaces and étale.*

Proof. Denote $\mathcal{C}$ the category introduced in the lemma. Observe that for $\varphi : U \to X$ in $\mathcal{C}$ the morphism φ is representable (by schemes) and affine, see Lemma 0AKN. Recall that $X_{affine,étale} = X_{red,affine,étale}$. Hence we can define a functor

$$\mathcal{C} \longrightarrow X_{affine,étale}, \quad (U \to X) \longmapsto U \times_X X_{red}$$

because $U \times_X X_{red}$ is an affine scheme.

To finish the proof we will construct a quasi-inverse. Namely, write $X = \operatorname{colim} X_\lambda$ as in Definition 0.1. For each λ we have $X_{red} \subset X_\lambda$ is a thickening. Thus for every λ we have an equivalence

$$X_{red,affine,étale} = X_{\lambda,affine,étale}$$

for example by More on Algebra, Lemma 0ALI. Hence if $U_{red} \to X_{red}$ is an étale morphism with U_{red} affine, then we obtain a system of étale morphisms $U_\lambda \to X_\lambda$ of affine schemes compatible with the transition morphisms in the system defining X. Hence we can take

$$U = \operatorname{colim} U_\lambda$$

as our affine formal algebraic space over X. The construction gives that $U \times_X X_\lambda = U_\lambda$. This shows that $U \to X$ is representable and étale. We omit the verification that the constructions are mutually inverse to each other. □